



SCHEME &SYLLABUS
BACHELOR'S OF SCIENCE (B.Sc. NEP)
(EFFECTIVE FROM JULY 2022)

Department of Science (UTD).
BHABHA UNIVERSITY, BHOPAL
NH-12, Hoshangabad Road, Jatkhedi, Bhopal-462026

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC 1st Semester

S.No	Course Code	Subject	CourseTitle	Credit		Theory		CCE/Internal				Practical/Project		
					Pre-Requisite (IfAny)	Max.	Min.	Class Test		Assignment/Presentation				Total Marks
								Max.	Min.	Max.	Min.	Max.	Min.	
SUBJECT I CORE COURSE /MAJOR SUBJECT (SELECT ANYONE SUBJECT)														
1	S1-BOTA1T	BOTANY	Applied Botany Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
	S1-BOTA1P	BOTANY PRACTICAL	Applied Botany Practical Paper-I	2	PCB	-	-	-		40	14	60	21	100
2	S1-CHEM1T	CHEMISTRY	Fundamental of Chemistry Paper-I	4	PCB/PCM	60	21	20	7	20	7	-	-	100
	S1-CHEM1P	CHEMISTRY PRACTICAL	Qualitative & Quantitative Chemical Analysis Practica lPaper-1	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S1-COSC1T	COMPUT SCIENCE	Computer System Architecture Paper-1	4	PCM	60	21	20	7	20	7	-	-	100
	S1-COSC1P	COMP SCIENCE PRACTICAL	Computer Architecture Lab Paper-1	2	PCM	-	-	-	-	40	14	60	21	100
4	S1-MATH1T	MATHEMATI CS	Algebra Vector Analysis And Geometry Paper-1	6	PCM	60	21	20	7	20	7	-	-	100
5	S1-MBIO1T	MICRO BIOLOGY	General Microbiology &Structure Paper-1	4	PCB	60	21	20	7	20	7	-	-	100
	S1-MBIO1P	MICRO BIOLOGY PRACTICAL	Study of Micro-Organism Practical Paper-1	2	PCB	-	-	--	-	40	14	60	21	100
6	S1-PHYS1T	PHYSICS	Thermodynamics &Statistical Physics Paper-1	4	PCM	60	21	20	7	20	7	-	-	100
	S1-PHYS1P	PHYSICS PRACTICAL	Thermodynamics &Statistical Physics Lab Paper-1	2	PCM	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC 1st Semester

7	S1-ZOOL1T	ZOOLOGY	Animal Diversity: Non-Chordata Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
	S1-ZOOL1P	ZOOLOGY PRACTICAL	Invertebrata Practical Paper-1	2	PCB	-	-	-	-	40	14	60	21	100
8	S1-BTEC1T	BIOTECHNOLOGY	Cell Biology and Biochemistry Paper-1	4	PCB	60	21	20	7	20	7	-	-	100
	S2-BTEC1P	BIOTECHNOLOGY PRACTICAL	Cell Biology and Biochemistry practical paper-1	2	PCB	-	-	-	-	40	14	60	21	100
SUBJECT II MINOR SUBJECT [SELECT ANY ONE SUBJECT OTHER THAN MAJOR]														
1	S1-BOTA1T	BOTANY	Applied Botany Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
	S1-BOTA1P	BOTANY PRACTICAL	Applied Botany Practical Paper-I	2	PCB	-	-	-	-	40	14	60	21	100
2	S1-CHEM1T	CHEMISTRY	Fundamental of Chemistry Paper-I	4	PCB/PCM	60	21	20	7	20	7	-	-	100
	S1-CHEM1P	CHEMISTRY PRACTICAL	Qualitative & Quantitative Chemical Analysis Practical Paper-1	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S1-COSC1T	COMPUTER SCIENCE	Computer System Architecture Paper-1	4	PCM	60	21	20	7	20	7	-	-	100
	S1-COSC1P	COMPUTER SCIENCE PRACTICAL	Computer Architecture Lab Paper-1	2	PCM	-	-	-	-	40	14	60	21	100
4	S1-MATH1T	MATHEMATICS	Algebra Vector Analysis And Geometry Paper-1	6	PCM	60	21	20	7	20	7	-	-	100
5	S1-MBIO1T	MICROBIOLOGY	General Microbiology & Structure Paper-1	4	PCB	60	21	20	7	20	7	-	-	100
	S1-MBIO1P	MICROBIOLOGY PRACTICAL	General Microbiology & Structure Paper-1	2	PCB	-	-	-	-	40	14	60	21	100
6	S1-PHYS1T	PHYSICS	Thermodynamics & Statistical Physics Paper-1	4	PCM	60	21	20	7	20	7	-	-	100
	S1-PHYS1P	PHYSICS PRACTICAL	Thermodynamics & Statistical Physics Lab Paper-1	2	PCM	-	-	-	-	40	14	60	21	100
7	S1-ZOOL1T	ZOOLOGY	Animal Diversity: Non-Chordata Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
	S1-ZOOL1P	ZOOLOGY PRACTICAL	Invertebrate Practical Paper-1	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

8	S1-BTEC1T	BIOTECHNOLOGY	Cell Biology and Biochemistry Paper-1	4	PCB	60	21	20	7	20	7	-	-	100
	S1-BTEC1P	BIOTECHNOLOGY PRACTICAL	Cell Biology and Biochemistry practical paper-	2	PCB	--	-	-	-	40	14	60	21	100

SUBJECT III GENERIC ELECTIVE COURSE [SELECT ANY ONE SUBJECT OTHER THAN MAJOR AND MINOR]														
1	S1-BOTA1T	BOTANY	Applied Botany Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
2	S1-CHEM1T	CHEMISTRY	Fundamental of Chemistry Paper-I	4	PCB/PCM	60	21	20	7	20	7	-	-	100
3	S1-COSC1T	COMPUTER SCIENCE	Computer System Architecture Paper-1	4	PCM	60	21	20	7	20	7	-	-	100
4	S1-MATH1T	MATHEMATICS	Algebra Vector Analysis And Geometry Paper-1	4	PCM	60	21	20	7	20	7	-	-	100
5	S1-MBIO1T	MICROBIOLOGY	General Microbiology & Structure Paper-1	4	PCB	60	21	20	7	20	7	-	-	100
6	S1-PHYS1T	PHYSICS	Thermodynamics & Statistical Physics Paper-1	4	PCM	60	21	20	7	20	7	-	-	100
7	S1-ZOOL1T	ZOOLOGY	Animal Diversity : Non-Chordata Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
8	S1-BTEC1T	BIOTECHNOLOGY	Cell Biology and Biochemistry Paper-1	4	PCB	60	21	20	7	20	7	-	-	100

ABILITY ENHANCEMENT COURSE[AEC]														
1	X1-FCEA1T	FOUNDATION COURSE	Hindi Language	2	Open For All	50	17	-	-	-	-	-	-	50
2	X1-FCHB1T	FOUNDATION COURSE	English Language and Indian Culture	2	Open For All	50	17	-	-	-	-	-	-	50
TOTAL CREDIT				20										600

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC IInd Semester

SUBJECT I CORE COURSE/MAJOR SUBJECT (The major subject that was selected in 1st semester is mandatory to be chosen in 2nd semester)														
1	S1-BOTA2T	BOTANY	Basic Botany Paper-II	4	PCB	60	21	20	7	20	7	-	-	100
	S1-BOTA2P	BOTANY PRACTICAL	Basic Botany Practical Paper-II	2	PCB	-	-	-	-	40	14	60	21	100
2	S1-CHEM2T	CHEMISTRY	Analytical Chemistry Paper-II	4	PCB/PCM	60	21	20	7	20	7	-	-	100
	S1-CHEM2P	CHEMISTRY PRACTICAL	Analytical Processes & Techniques Paper-II	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S1-COSC2T	COMPSCIENCE	Programming Methodology & Data Structure Paper-II	4	PCM	60	21	20	7	20	7	-	-	100
	S1-COSC2P	COMPSCIENCE PRACTICAL	Office Tools & Programming Methodology Lab Paper-II	2	PCM	-	-	-	-	40	14	60	21	100
4	S1-MATH2T	MATHEMATICS	Calculus And Differential Equations Paper-II	6	PCM	60	21	20	7	20	7	-	-	100
5	S1-MBIO2T	MICROBIOLOGY	Microbial Techniques Paper-II	4	PCB	60	21	20	7	20	7	-	-	100
	S1-MBIO2P	MICROBIOLOGY PRACTICAL	Microbial Tools & Techniques Paper-II	2	PCB	-	-	-	-	40	14	60	21	100
6	S1-PHYS2T	PHYSICS	Mechanics & General Properties of Matter Paper-II	4	PCM	60	21	20	7	20	7	-	-	100
	S1-PHYS2P	PHYSICS PRACTICAL	Mechanics & General Properties of Matter Lab Paper-II	2	PCM	-	-	-	-	40	14	60	21	100
7	S1-ZOOL2T	ZOOLOGY	Cell biology ,Reproductive Biology and Developmental Biology Paper-II	4	PCB	60	21	20	7	20	7	-	-	100
	S1-ZOOL2P	ZOOLOGY PRACTICAL	Cytology ,Reproductive Biology and Embryology Paper-II	2	PCB	-	-	-	-	40	14	60	21	100
8	S1-BTEC2T	BIOTECHNOLOGY	Microbiology and Immunology-II	4	PCB	60	21	20	7	20	7	-	-	100
	S1-BTEC2P	BIOTECHNOLOGY PRACTICAL	Microbiology and Immunology-II	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC IInd Semester

SUBJECT II MINOR SUBJECT (The minor subject that was selected in 1st semester is mandatory to be chosen in 2nd semester)														
1	S1-BOTA2T	BOTANY	Basic Botany Paper-II	4	PCB	60	21	20	7	20	7	-	-	100
	S1-BOTA2P	BOTANY PRACTICAL	Basic Botany Practical Paper-II	2	PCB	-	-	-	-	40	14	60	21	100
2	S1-CHEM2T	CHEMISTRY	Analytical Chemistry Paper-II	4	PCB/PCM	60	21	20	7	20	7	-	-	100
	S1-CHEM2P	CHEMISTRY PRACTICAL	Analytical Processes & Techniques Paper-II	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S1-COSC2T	COMPUTER SCIENCE	Programming Methodology & Data Structure Paper-II	4	PCM	60	21	20	7	20	7	-	-	100
	S1-COSC2P	COMPUTER SCIENCE PRACTICAL	Office Tools & Programming Methodology Lab Paper-II	2	PCM	-	-	-	-	40	14	60	21	100
4	S1-MATH2T	MATHEMATICS	Calculus And Differential Equations Paper-II	4	PCM	60	21	20	7	20	7	-	-	100
5	S1-MBIO2T	MICROBIOLOGY	Microbial Techniques Paper-II	4	PCB	60	21	20	7	20	7	-	-	100
	S1-MBIO2P	MICROBIOLOGY PRACTICAL	Microbial Tools & Techniques Paper-II	2	PCB	-	-	-	-	40	14	60	21	100
6	S1-PHYS2T	PHYSICS	Mechanics & General Properties of Matter Paper-II	4	PCM	60	21	20	7	20	7	-	-	100
	S1-PHYS2P	PHYSICS PRACTICAL	Mechanics & General Properties of Matter Lab Paper-II	2	PCM	-	-	-	-	40	14	60	21	100
7	S1-ZOOL2T	ZOOLOGY	Cell biology, Reproductive Biology and Developmental Biology Paper-II	4	PCB	60	21	20	7	20	7	-	-	100
	S1-ZOOL2P	ZOOLOGY PRACTICAL	Cytology, Reproductive Biology and Embryology Paper-II--	2	PCB	-	-	-	-	40	14	60	21	100
8	S1-BTEC2T	BIOTECHNOLOGY	Microbiology and Immunology-II	4	PCB	60	21	20	7	20	7	-	-	100
	S1-BTEC2p	BIOTECHNOLOGY Practical	Microbiology and Immunology-II	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

SUBJECT II GENERIC ELECTIVE COURSE (The Generic Elective Course that was selected in 1st semester is mandatory to be chosen in 2nd semester)														
1	S1-BOTA2T	BOTANY	Basic Botany Paper-II	4	PCB	60	21	20	7	20	7	-	-	100
2	S1-CHEM2T	CHEMISTRY	Analytical Chemistry Paper-II	4	PCB/PCM	60	21	20	7	20	7	-	-	100
3	S1-COSC2T	COMP SCIENCE	Programming Methodology & Data Structure Paper-II	4	PCM	60	21	20	7	20	7	-	-	100
4	S1-MATH2T	MATHEMATICS	Calculus And Differential Equations Paper-II	4	PCM	60	21	20	7	20	7	-	-	100
5	S1-MBIO2T	MICROBIOLOGY	Microbial Techniques Paper-II	4	PCB	60	21	20	7	20	7	-	-	100
6	S1-PHYS2T	PHYSICS	Mechanics & General Properties of Matter Paper-II	4	PCM	60	21	20	7	20	7	-	-	100
7	S1-ZOOL2T	ZOOLOGY	Cell biology ,Reproductive Biology and Developmental Biology Paper-II	4	PCB	60	21	20	7	20	7	-	-	100
8	S1-BTEC2T	BIOTECHNOLOGY	Microbiology and Immunology-II	4	PCB	60	21	20	7	20	7	-	-	100

ABILITY ENHANCEMENT COURSE														
1	X1-FCAC1T	FOUNDATION COURSE	Environmental Education	2	Open For All	50	17	-	-	-	-	-	-	50
2	AI-YOSC1F	FOUNDATION COURSE	Yoga And Mediation	2	Open For All	50	17	-	-	-	-	-	-	50
TOTAL CREDIT				20										600

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC IIIrd Semester

SUBJECT I CORECOURSE/MAJOR SUBJECT (The major subject that was selected in 1st semester is mandatory to be chosen in 3rd semester)														
1	S2-BOTA1T	BOTANY	Plant Anatomy and Embryology paper 1	4	PCB	60	21	20	7	20	7	-	-	100
	S2-BOTA1P	BOTANY PRACTICAL	Plant Anatomy and Embryology ,practical paper1	2	PCB	-	-		-	40	14	60	21	100
2	S2-CHEM1T	CHEMISTRY	Reaction, Reagents and Mechanism in Organic Chemistry Paper-I	4	PCB/PC M	60	21	20	7	20	7	-	-	100
	S2-CHEM1P	CHEMISTRY PRACTICAL	Organic Qualitative Analysis, Reaction and Synthesis Paper-I	2	PCB/PC M	-	-	-	-	40	14	60	21	100
3	S2-COSC1T	COMPSC IENCE	Computer Net work& Information Security. Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
	S2-COSC1P	COMPSCIEN CE PRACTICAL	Computer Network Lab Paper-I	2	PCM	-	-	-	-	40	14	60	21	100
4	S2-MATH1T	MATHEMATICS	Abstract Algebra and Linear Algebra Paper-I	6	PCM	60	21	20	7	20	7	-	-	100
5	S2-MBIO1T	MICROBIOLOGY	Microbial Physiology and Metabolism Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
	S2-MBIO1P	MICROBIOLOGY PRACTICAL	Microbial Biochemistry and Metabolism Paper-I	2	PCB	-	-	-	-	40	14	60	21	100
6	S2-PHYS1T	PHYSICS	Waves and Optics Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
	S2-PHYS1P	PHYSICS PRACTICAL	Waves and Optics Lab Paper-I	2	PCM	-	-	-	-	40	14	60	21	100
7	S2-ZOOL1T	ZOOLOGY	Diversity of Chordates and Comparative Anatomy Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
	S2-ZOOL1P	ZOOLOGY PRACTICAL	Chordate Zoology Paper-I	2	PCB	-	-	-	-	40	14	60	21	100
8	S2-BTEC1T	BIOTECHNOLOGY	Basic Molecular Biology -I	4	PCB	60	21	20	7	20	7	-	-	100
	S2-BTEC1P	BIOTECHNOLOGY PRACTICAL	Labwork for Basic Molecular Biology -I	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC IIIrd Semester

SUBJECT II MINOR SUBJECT (The minor subject that was selected in 1st semester is mandatory to be chosen in 3rd semester)														
1	S2-BOTA1T	BOTANY	Plant Anatomy and Embryology paper 1	4	PCB	60	21	20	7	20	7	-	-	100
	S2-BOTA1P	BOTANY PRACTICAL	Plant Anatomy and Embryology ,practical paper1	2	PCB	-	-	-	-	40	14	60	21	100
2	S2-CHEM1T	CHEMISTRY	Reaction, Reagents and Mechanism in Organic Chemistry Paper-I	4	PCB/PCM	60	21	20	7	20	7	-	-	100
	S2-CHEM1P	CHEMISTRY PRACTICAL	Organic Qualitative Analysis, Reaction and Synthesis Paper-I	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S2-COSC1T	COMPUTER SCIENCE	Computer Net work& Information Security. Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
	S2-COSC1P	COMPUTER SCIENCE PRACTICAL	Computer Network Lab Paper-I	2	PCM	-	-	-	-	40	14	60	21	100
4	S2-MATH1T	MATHEMATICS	Abstract Algebra and Linear Algebra Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
5	S2-MBIO1T	MICROBIOLOGY	Microbial Physiology and Metabolism Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
	S2-MBIO1P	MICROBIOLOGY PRACTICAL	Microbial Biochemistry and Metabolism Paper-I	2	PCB	-	-	-	-	40	14	60	21	100
6	S2-PHYS1T	PHYSICS	Waves and Optics Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
	S2-PHYS1P	PHYSICS PRACTICAL	Waves and Optics Lab Paper-I	2	PCM	-	-	-	-	40	14	60	21	100
7	S2-ZOOL1T	ZOOLOGY	Diversity of Chordates and Comparative Anatomy Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
	S2-ZOOL1P	ZOOLOGY PRACTICAL	Chordate Zoology Paper-I	2	PCB	-	-	-	-	40	14	60	21	100
8	S2-BTEC1T	BIOTECHNOLOGY	Basic Molecular Biology -I	4	PCB	60	21	20	7	20	7	-	-	100
	S2-BTEC1P	BIOTECHNOLOGY Practical	Lab work for Basic Molecular Biology -I	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC IIIrd Semester

Subject III GENERIC ELECTIVE COURSE (The Generic Elective Course that was selected in 1st semester is mandatory to be chosen in 3rd semester)														
1	S2-BOTA1T	BOTANY	Plant Anatomy and Embryology paper 1	4	PCB	60	21	20	7	20	7	-	-	100
2	S2-CHEM1T	CHEMISTRY	Reaction, Reagents and Mechanism in Organic Chemistry Paper-I	4	PCB/PCM	60	21	20	7	20	7	-	-	100
3	S2-COSC1T	COMPUTER SCIENCE	Computer Network & Information Security Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
4	S2-MATH1T	MATHEMATICS	Abstract Algebra and Linear Algebra Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
5	S2-MBIO1T	MICROBIOLOGY	Microbial Physiology and Metabolism Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
6	S2-PHYS1T	PHYSICS	Waves and Optics Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
7	S2-ZOOL1T	ZOOLOGY	Diversity of Chordates and Comparative Anatomy Paper-I	4	PCB	60	21	20	7	20	7	-	-	100
8	S2-BTEC1T	BIOTECHNOLOGY	Basic Molecular Biology -I	4	PCB	60	21	20	7	20	7	-	-	100

Skill ENHANCEMENT COURSE (SEC) / Vocational Course														
1	V2-PSY-DEVT	Vocational Course	Personality Development	2	Open For All	50	17	-	-	-	-	-	-	50
2	V2-PSY-DEVP	Vocational Course	Practical	2	Open For All	50	17	-	-	-	-	-	-	50
TOTAL CREDIT				20										600

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC IVth Semester

SUBJECT I CORE COURSE/MAJOR SUBJECT (The major subject that was selected in 1st semester is mandatory to be chosen in 4th semester)														
1	S2-BOTA2T	BOTANY	Industrial Botany paper 2	4	PCB	60	21	20	7	20	7	-	-	100
	S2-BOTA2P	BOTANY PRACTICAL	Industrial Botany ,practical paper2	2	PCB	-	-	-	-	40	14	60	21	100
2	S2-CHEM2T	CHEMISTRY	Transition Elements, Chemi- energetics, phase Equilibriya Paper-2	4	PCB/PC M	60	21	20	7	20	7	-	-	100
	S2-CHEM2P	CHEMISTRY PRACTICAL	Metal Complex Preparation , Thermo chemistry & Phase equilibriya experiments Paper-2	2	PCB/PC M	-	-	-	-	40	14	60	21	100
3	S2-COSC2T	COMPSC IENCE	Object Oriented Programing with java.Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
	S2-COSC2P	COMPSCIEN CE PRACTICAL	Java Programing Lab Paper-2	2	PCM	-	-	-	-	40	14	60	21	100
4	S2-MATH2T	MATHEMATICS	Advanced calculus and partial Differential Equation Paper-2	6	PCM	60	21	20	7	20	7	-	-	100
5	S2-MBIO2T	MICROBIOLOGY	Microbial Diversity and Growth Paper-2	4	PCB	60	21	20	7	20	7	-	-	100
	S2-MBIO2P	MICROBIOLOGY PRACTICAL	Isolation and Growth study of microorganism Paper-2	2	PCB	-	-	-	-	40	14	60	21	100
6	S2-PHYS2T	PHYSICS	Electricity Magnetism and Electro magnetics Theory Paper-2	4	PCM	60	21	20	7	20	7	-	-	100
	S2-PHYS2P	PHYSICS PRACTICAL	Electricity Magnetism and EMT Lab Paper-2	2	PCM	-	-	-	-	40	14	60	21	100
7	S2-ZOOL2T	ZOOLOGY	Physiology and Biochemistry Paper-2	4	PCB	60	21	20	7	20	7	-	-	100
	S2-ZOOL2P	ZOOLOGY PRACTICAL	System Physiology and Biochemistry Paper-2	2	PCB	-	-	-	-	40	14	60	21	100
8	S2-BTEC2T	BIOTECHNOLOGY	Recombinant DNA Technology paper-2	4	PCB	60	21	20	7	20	7	-	-	100
	S2-BTEC2P	BIOTECHNOLOGY PRACTICAL	Lab work for Recombinant DNA Technology-paper2	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC IVth Semester

SUBJECT II MINOR SUBJECT (The minor subject that was selected in 1st semester is mandatory to be chosen in 3rd semester)														
1	S2-BOTA2T	BOTANY	Industrial Botany paper 2	4	PCB	60	21	20	7	20	7	-	-	100
	S2-BOTA2P	BOTANY PRACTICAL	Industrial Botany ,practical paper2	2	PCB	-	-	-	-	40	14	60	21	100
2	S2-CHEM2T	CHEMISTRY	Transition Elements, Chemi- energetics, phase Equilibria Paper-2	4	PCB/PCM	60	21	20	7	20	7	-	-	100
	S2-CHEM2P	CHEMISTRY PRACTICAL	Metal Complex Preparation , Thermochemistry & Phase equilibria experiments Paper-2	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S2-COSC2T	COMPUTER SCIENCE	Object Oriented Programing with java.Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
	S2-COSC2P	COMPUTER SCIENCE PRACTICAL	Java Programing Lab Paper-2	2	PCM	-	-	-	-	40	14	60	21	100
4	S2-MATH2T	MATHEMATICS	Advanced calculus and partial Differential Equation Paper-2	4	PCM	60	21	20	7	20	7	-	-	100
5	S2-MBIO2T	MICROBIOLOGY	Microbial Diversity and Growth Paper-2	4	PCB	60	21	20	7	20	7	-	-	100
	S2-MBIO2P	MICROBIOLOGY PRACTICAL	Isolation and Growth study of microorganism Paper-2	2	PCB	-	-	-	-	40	14	60	21	100
6	S2-PHYS2T	PHYSICS	Electricity Magnetism and Electromagnetics Theory Paper-2	4	PCM	60	21	20	7	20	7	-	-	100
	S2-PHYS2P	PHYSICS PRACTICAL	Electricity Magnetism and EMT Lab Paper2	2	PCM	-	-	-	-	40	14	60	21	100
7	S2-ZOOL2T	ZOOLOGY	Physiology and Biochemistry Paper-2	4	PCB	60	21	20	7	20	7	-	-	100
	S2-ZOOL2p	ZOOLOGY PRACTICAL	System Physiology and Biochemistry Paper-2	2	PCB	-	-	-	-	40	14	60	21	100
8	S2-BTEC2T	BIOTECHNOLOGY	Recombinant DNA Technology paper-2	4	PCB	60	21	20	7	20	7	-	-	100
	S2-BTEC2P	BIOTECHNOLOGY Practical	Labwork for Recombinant DNA Technology –paper2	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC IVth Semester

Subject III GENERIC ELECTIVE COURSE (The Generic Elective Course that was selected in 1st semester is mandatory to be chosen in IVth semester)														
1	S2-BOTA2T	BOTANY	Industrial Botany paper 2	4	PCB	60	21	20	7	20	7	-	-	100
2	S2-CHEM2T	CHEMISTRY	Transition Elements, Chemi- energetics, phase Equilibria Paper-2	4	PCB/PCM	60	21	20	7	20	7	-	-	100
3	S2-COSC2T	COMPSCIENCE	Object Oriented Programing with java.Paper-I	4	PCM	60	21	20	7	20	7	-	-	100
4	S2-MATH2T	MATHEMATICS	Advanced calculus and partial Differential Equation Paper-2	4	PCM	60	21	20	7	20	7	-	-	100
5	S2-MBIO2T	MICROBIOLOGY	Microbial Diversity and Growth Paper-2	4	PCB	60	21	20	7	20	7	-	-	100
6	S2-PHYS2T	PHYSICS	Electricity Magnetism and Electromagnetics Theory Paper-2	4	PCM	60	21	20	7	20	7	-	-	100
7	S2-ZOOL2T	ZOOLOGY	Physiology and Biochemistry Paper-2	4	PCB	60	21	20	7	20	7	-	-	100
8	S2-BTEC2T	BIOTECHNOLOGY	Recombinant DNA Technology paper-2	4	PCB	60	21	20	7	20	7	-	-	100

Skill ENHANCEMENT COURSE(SEC) /Vocational Course)														
1	V2-COM-DIGT-DEVT	Vocational Course	E-COMMERCE	4	Open For All	60	21	20	7	20	7	-	-	100
								-	-	-	-	-	-	
TOTAL CREDIT				20										600

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC VthSemester

SUBJECT I CORECOURSE/MAJORSUBJECT (The major subject that was selected in 1st semester is mandatory to be chosen in 5th semester)														
1	S3-BOTA1T	BOTANY	Plant Physiology and metabolism	4	PCB	60	21	20	7	20	7	-	-	100
	S3-BOTA1P	BOTANY PRACTICAL	Plant Physiology and metabolism	2	PCB	-	-	-	-	40	14	60	21	100
2	S3-CHEM1T	CHEMISTRY	Green and Chemistry Agriculture	4	PCB/PCM	60	21	20	7	20	7	-	-	100
	S3-CHEM1P	CHEMISTRY PRACTICAL	Green and Chemistry Agriculture	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S3-COSC1T	COMPS CIENCE	Operating system Paper	4	PCM	60	21	20	7	20	7	-	-	100
	S3-COSC1P	COMPSCIEN CE PRACTICAL	Operating system Lab	2	PCM	-	-	-	-	40	14	60	21	100
4	S3-MATH1T	MATHEMATICS	Numwrical methods and scientific computation	6	PCM	60	21	20	7	20	7	-	-	100
5	S3-MBIO1T	MICROBIOLOGY	Microbial Genetics and molecular Biology	4	PCB	60	21	20	7	20	7	-	-	100
	S3-MBIO1P	MICROBIOLOGY PRACTICAL	Microbial Genetics and molecular Biology	2	PCB	-	-	-	-	40	14	60	21	100
6	S3-PHYS1T	PHYSICS	Quantum Atomic and molecular Physics	4	PCM	60	21	20	7	20	7	-	-	100
	S3-PHYS1P	PHYSICS PRACTICAL	Quantum Atomic and molecular Physics	2	PCM	-	-	-	-	40	14	60	21	100
77	S3-ZOOL1T	ZOOLOGY	Aquaculture	4	PCB	60	21	20	7	20	7	-	-	100
	S3-ZOOL1P	ZOOLOGY PRACTICAL	Applied Aquaculture	2	PCB	-	-	-	-	40	14	60	21	100
8	S3-BTEC1T	BIOTECHNOLOGY	Industrial Biotechnology	4	PCB	60	21	20	7	20	7	-	-	100
	S3-BTEC1P	BIOTECHNOLOGY PRACTICAL	Industrial Biotechnology	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC Vth Semester

SUBJECT II DSE-1

Note: The Major subject that was in the 1st semester is compulsory to select as a DSE 1(Disciplinary specific elective) subject in the 5th semester

1	S3-BOTA1D	BOTANY	Ecology and forestry	2	PCB	60	21	20	7	20	7	-	-	100
	S3-BOTA1Q	BOTANY PRACTICAL	Ecology and forestry	2	PCB	-	-	-	-	40	14	60	21	100
2	S3-CHEM1D	CHEMISTRY	Laboratory skill techniques & management	2	PCB/PCM	60	21	20	7	20	7	-	-	100
	S3-CHEM1Q	CHEMISTRY PRACTICAL	Exercises for Development of lab skills	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S3-COSC1D	COMPS CIENCE	Programming with python	2	PCM	60	21	20	7	20	7	-	-	100
	S3-COSC1Q	COMP SCIENCE PRACTICAL	Python programming lab	2	PCM	-	-	-	-	40	14	60	21	100
4	S3-MATH1D	MATHEMATICS	Elements of Discrete mathematics	2	PCM	60	21	20	7	20	7	-	-	100
5	S3-MBIO1D	MICROBIOLOGY	Food Microbiology	2	PCB	60	21	20	7	20	7	-	-	100
	S3-MBIO1Q	MICROBIOLOGY PRACTICAL	Techniques of food Microbiology	2	PCB	-	-	-	-	40	14	60	21	100
6	S3-PHYS1D	PHYSICS	Solid state physics and electronics	2	PCM	60	21	20	7	20	7	-	-	100
	S3-PHYS1Q	PHYSICS PRACTICAL	Solid state physics and electronics	2	PCM	-	-	-	-	40	14	60	21	100
7	S3-ZOOL1D	ZOOLOGY	Wild life conservation and Management	2	PCB	60	21	20	7	20	7	-	-	100
	S3-ZOOL1Q	ZOOLOGY PRACTICAL	Importance of wild life	2	PCB	-	-	-	-	40	14	60	21	100
8	S3-BTEC1D	BIOTECHNOLOGY	Agriculture Biotechnology	2	PCB	60	21	20	7	20	7	-	-	100
	S3-BTEC1Q	BIOTECHNOLOGY PRACTICAL	Agriculture Biotechnology	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

Skill ENHANCEMENTCOURSE(SEC) /Vocational Course)														
1	V3-COM-REMT	Vocational Course	Consumer Affairs and customer care	4	Open For All	60	21	20	7	20	7	-	-	100

Field Project/Internship/apprenticeship														
1				6	Open For All	60	21	20	7	20	7	-	-	100
								-	-	-	-	-	-	
TOTALCREDIT				20										600

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC VIth Semester

SUBJECT I CORECOURSE/MAJORSUBJECT (The major subject that was selected in 1st semester is mandatory to be chosen in 6th semester)														
1	S3-BOTA2T	BOTANY	Ethnobotany	4	PCB	60	21	20	7	20	7	-	-	100
	S3-BOTA2P	BOTANY PRACTICAL	Ethnobotany	2	PCB	-	-	-	-	40	14	60	21	100
2	S3-CHEM2T	CHEMISTRY	Pharmaceutical and medicinal chemistry	4	PCB/PCM	60	21	20	7	20	7	-	-	100
	S3-CHEM2P	CHEMISTRY PRACTICAL	Pharmaceutical and medicinal chemistry	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S3-COSC2T	COMPSCIENCE	Data Analysis and visualization with python	4	PCM	60	21	20	7	20	7	-	-	100
	S3-COSC2P	COMPSCIENCE PRACTICAL	Data Analysis and visualization with python	2	PCM	-	-	-	-	40	14	60	21	100
4	S3-MATH2T	MATHEMATICS	Fundamental of Boolean Algebra	6	PCM	60	21	20	7	20	7	-	-	100
5	S3-MBIO2T	MICROBIOLOGY	Environmental Microbiology	4	PCB	60	21	20	7	20	7	-	-	100
	S3-MBIO2P	MICROBIOLOGY PRACTICAL	Environmental Microbiology	2	PCB	-	-	-	-	40	14	60	21	100
6	S3-PHYS2T	PHYSICS	Quantum Mechanics ,solid state physics, and Devices	4	PCM	60	21	20	7	20	7	-	-	100
	S3-PHYS2P	PHYSICS PRACTICAL	Quantum Mechanics ,solid state physics, and Devices	2	PCM	-	-	-	-	40	14	60	21	100
7	S3-ZOOL2T	ZOOLOGY	Genetics	4	PCB	60	21	20	7	20	7	-	-	100
	S3-ZOOL2P	ZOOLOGY PRACTICAL	Experimental Genetics	2	PCB	-	-	-	-	40	14	60	21	100
8	S3-BTEC2T	BIOTECHNOLOGY	Applied Biotechnology	4	PCB	60	21	20	7	20	7	-	-	100
	S3-BTEC2P	BIOTECHNOLOGY PRACTICAL	Applied Biotechnology	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC VIth Semester

SUBJECT II DSE-2														
Note: The Major subject that was in the 1 st semester is compulsory to select as a DSE- II (Disciplinary specific elective) subject in the 6 th semester														
1	S3-BOTA2D	BOTANY	Cytology, Plant Breeding and Biotechnology	2	PCB	60	21	20	7	20	7	-	-	100
	S3-BOTA2Q	BOTANY PRACTICAL	Cytology, Plant Breeding and Biotechnology	2	PCB	-	-	-	-	40	14	60	21	100
2	S3-CHEM2D	CHEMISTRY	Instrumental Techniques in chemistry	2	PCB/PCM	60	21	20	7	20	7	-	-	100
	S3-CHEM2Q	CHEMISTRY PRACTICAL	Instrumental Analytical Techniques in chemistry	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S3-COSC2D	COMPS CIENCE	PHP & my SQL	2	PCM	60	21	20	7	20	7	-	-	100
	S3-COSC2Q	COMP SCIENCE PRACTICAL	PHP & my SQL LAB	2	PCM	-	-	-	-	40	14	60	21	100
4	S3-MATH2D	MATHEMATICS	Probability and statistics	2	PCM	60	21	20	7	20	7	-	-	100
5	S3-MBIO2D	MICROBIOLOGY	Industrial Microbiology	2	PCB	60	21	20	7	20	7	-	-	100
	S3-MBIO2Q	MICROBIOLOGY PRACTICAL	Method of Microbial Production	2	PCB	-	-	-	-	40	14	60	21	100
6	S3-PHYS2D	PHYSICS	Astronomy and space physics	2	PCM	60	21	20	7	20	7	-	-	100
	S3-PHYS2Q	PHYSICS PRACTICAL	Astronomy and space physics	2	PCM	-	-	-	-	40	14	60	21	100
7	S3-ZOOL2D	ZOOLOGY	Insect Taxonomy and Applied Entomology	2	PCB	60	21	20	7	20	7	-	-	100
	S3-ZOOL2Q	ZOOLOGY PRACTICAL	Applied Entomology	2	PCB	-	-	-	-	40	14	60	21	100
8	S3-BTEC2D	BIOTECHNOLOGY	Environmental Biotechnology	2	PCB	60	21	20	7	20	7	-	-	100
	S3-BTEC2Q	BIOTECHNOLOGY PRACTICAL	Environmental Biotechnology	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

DEPARTMENT OF SCIENCE SCHEME OF BSC VIth Semester

SUBJECT III, DSE-3														
Note: The Major subject that was in the 1 st semester is compulsory to select as a DSE- III (Disciplinary specific elective) subject in the 6 th semester														
1	S3-BOTA3D	BOTANY	Biodiversity and Economic Botany	2	PCB	60	21	20	7	20	7	-	-	100
	S3-BOTA3Q	BOTANY PRACTICAL	Biodiversity and Economic Botany	2	PCB	-	-	-	-	40	14	60	21	100
2	S3-CHEM3D	CHEMISTRY	Bio physical , Bio inorganic and organometallic Chemistry	2	PCB/PCM	60	21	20	7	20	7	-	-	100
	S3-CHEM3Q	CHEMISTRY PRACTICAL	Synthesis and Analytical Techniques	2	PCB/PCM	-	-	-	-	40	14	60	21	100
3	S3-COSC3D	COMPUTER SCIENCE	Cloud Computing	2	PCM	60	21	20	7	20	7	-	-	100
	S3-COSC3Q	COMPUTER SCIENCE PRACTICAL	Cloud Computing Lab	2	PCM	-	-	-	-	40	14	60	21	100
4	S3-MATH3D	MATHEMATICS	Integral transform	2	PCM	60	21	20	7	20	7	-	-	100
5	S3-MBIO3D	MICROBIOLOGY	Microbes as Pathogen	2	PCB	60	21	20	7	20	7	-	-	100
	S3-MBIO3Q	MICROBIOLOGY PRACTICAL	Microbes as Pathogen	2	PCB	-	-	-	-	40	14	60	21	100
6	S3-PHYS3D	PHYSICS	Nuclear and Particle Physics	2	PCM	60	21	20	7	20	7	-	-	100
	S3-PHYS3Q	PHYSICS PRACTICAL	Nuclear and Particle Physics	2	PCM	-	-	-	-	40	14	60	21	100
7	S3-ZOOL3D	ZOOLOGY	Ecology , Biodiversity and Evolution	2	PCB	60	21	20	7	20	7	-	-	100
	S3-ZOOL3Q	ZOOLOGY PRACTICAL	Environmental Biology	2	PCB	-	-	-	-	40	14	60	21	100
8	S3-BTEC3D	BIOTECHNOLOGY	Bioinformatics	2	PCB	60	21	20	7	20	7	-	-	100
	S3-BTEC3Q	BIOTECHNOLOGY PRACTICAL	Bioinformatics	2	PCB	-	-	-	-	40	14	60	21	100

BHABHA UNIVERSITY BHOPAL(M.P.)

Field Project/Internship/apprenticeship														
1			Field Project/Internship/apprenticeship	6	Open For All	60	21	20	7	20	7	-	-	100
								-	-	-	-	-	-	
TOTALCREDIT				20										600

